

# Memory and chaos in a physics-constrained relaxation substrate: phase diagram, multi-timescale forgetting, and disturbance robustness

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## Abstract

Reservoir computing exploits the fading-memory dynamics of a physical substrate, yet the memory–chaos trade-off is usually studied on abstract recurrent networks with hand-tuned leak rates. We characterize a physics-constrained relaxation substrate inspired by  $\text{Si}_3\text{N}_4$  shallow-trap charge storage:  $N$  units with a log-normal time-constant spectrum (median  $\tau_0 \approx 174 \mu\text{s}$ , coefficient of variation  $\text{CV} = 0.20$ ) coupled through a per-pulse, topology-dependent modulation of the injection coefficient. Sweeping the coupling strength  $\kappa$  across five topology families (three negative-feedback families undergo the transition described below) and measuring the finite-time Lyapunov exponent  $\lambda$ , the held-out Jaeger memory capacity, and input separation, we find a sharp order–chaos transition at  $\kappa^* \in (25, 30)$ : the held-out memory capacity peaks 24–53% above the uncoupled baseline just before the transition, and deep chaos destroys memory. The decay of linear memory follows an analytically derived forgetting kernel  $M(t) = \int p(\tau)e^{-t/\tau} d\tau$  over the log-normal trap spectrum (Pearson  $r = 0.97$  against the measured memory-capacity curve). The  $1/e$  horizon stays near  $\tau_0/\bar{\Delta}t \approx 16$  pulses as the spectrum widens (numerically 12–16 pulses for  $\text{CV} \in [0.02, 1.0]$ ), while the width  $\text{CV}$  controls the tail weight. Finally, a homeostatic regulator that estimates  $\lambda$  online and adjusts  $\kappa$  to a near-critical target improves post-disturbance held-out memory by 7.9–18% under temperature drift, edge pruning, and readout noise.

**Keywords:** chaos; reservoir computing; memory capacity; forgetting kernel; homeostasis; relaxation substrate.

## 1 Introduction

Physical reservoir computing proposes that a complex dynamical substrate — a spiking neural microcircuit [1], an optoelectronic system [2], a memristive array [3], or a charge-trap device [4] — can serve as a nonlinear, fading-memory kernel whose instantaneous state is linearly read out by a trained network. Physical implementations now span optoelectronic delay systems [2, 5], photonic integrated circuits [6], a single nonlinear node under time multiplexing [7], and neuromorphic device platforms [8, 9]; see [10, 11, 12] for reviews. The theoretical foundations are well established: the fading-memory property dates to the Volterra/Wiener system-identification literature [13], and echo state networks [14, 15] and liquid state machines [1] rely on the *echo state property* (contracting, input-forgetting dynamics), and the *edge of chaos* maximizes the computational capability of driven recurrent systems [16, 17, 18], building on the earlier characterization of the onset of chaos in

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random recurrent networks [19]. Dambre et al. [20] formalized the total information-processing capacity of dynamical systems, showing a universal trade-off between linear memory and nonlinear processing. On the memory side, multi-timescale synaptic models [21, 22] show that memory retention is dramatically extended by a *spectrum* of time constants: cascade models achieve near power-law forgetting from many exponentially decaying components.

For a physical reservoir, the time-constant spectrum is not a free hyperparameter — it is a *material property*. Shallow-trap charge storage in  $\text{Si}_3\text{N}_4$  exhibits thermally activated relaxation with a log-normal spread of trap time constants; our prior work [4] established that this substrate encodes pulse-interval patterns through exponential relaxation with  $\tau_0 \approx 174 \mu\text{s}$  at 300 K and a device-to-device spread of  $\text{CV} = 0.20$ , achieving 79.6% four-class hybrid encoding accuracy in a purely numerical setting (Digital-Model: the calibrated device simulator, with no fabricated hardware in the loop; all experiments in this paper are at this same fidelity level). That work, however, treated the substrate as a *parallel* array of independent devices; the coupling between devices — the element that could push the substrate toward richer, near-critical dynamics — was only implemented as a quasi-static modulation of the effective injection coefficient across blocks.

This paper closes that gap. We introduce the *per-pulse temporal generalization* of the topology coupling: every pulse, each unit’s injection coefficient is modulated by a topology-dependent contrast against its neighbors’ current ratios, making the substrate a genuinely recurrent, tunable dynamical system with a physical chaos knob  $\kappa$ . Our contributions are:

1. **A memory–chaos phase diagram** for a physics-constrained relaxation substrate: the finite-time Lyapunov exponent  $\lambda$ , the held-out Jaeger memory capacity, and the input separation, all as functions of the coupling strength  $\kappa$ , across five topology families (bidirectional ring, unidirectional ring, hub\_star, lateral-inhibition ring, random graph) and two coupling sign conventions (negative- and positive-feedback contrast), with an additive state-coupling control.
2. **An analytic forgetting kernel**  $M(t) = \int p(\tau)e^{-t/\tau} d\tau$  for the log-normal trap spectrum, validated against the measured memory-capacity curve (Pearson  $r = 0.97$ ), which turns the device-physics spread CV into a *design knob for the forgetting curve*.
3. **A homeostatic chaos regulator** that estimates  $\lambda$  online from Benettin twin trajectories and adjusts  $\kappa$  to a near-critical target, restoring 7.9–18% of held-out memory after environmental disturbances (temperature drift, edge pruning, readout noise) where fixed coupling cannot.

Throughout, we use a *held-out* memory-capacity protocol: ridge fits are trained on the first 70% of a driving stream and evaluated on the last 30% (with a  $k_{\max}$ -length leakage buffer). We show that the common in-sample Jaeger convention dramatically overstates memory in the chaotic regime, which is a methodological caution for the field.

## 2 Substrate model

### 2.1 Single-unit relaxation

Each unit models a  $\text{Si}_3\text{N}_4$  shallow-trap population — trap-controlled charge transport of this kind is documented independently in  $\text{SiN}_x$  memristors [23] — with effective trap occupancy  $x_i \in [0, 1]$ . A write pulse of width  $p_w = 1 \mu\text{s}$  injects charge with coefficient  $\alpha$ ; the occupancy then relaxes exponentially with the unit’s trap time constant  $\tau_i$  both during the pulse width and during the inter-pulse interval  $\Delta t_t$ . The per-pulse update is

$$x_i(t+1) = \left[ x_i(t) + \alpha_{\text{eff},i}(t)(1 - x_i(t)) \right] e^{-p_w/\tau_i} e^{-\Delta t_t/\tau_i}, \quad x_i \in [0, 1], \quad (1)$$

Fig 1 (Paper A): physics-constrained relaxation substrate

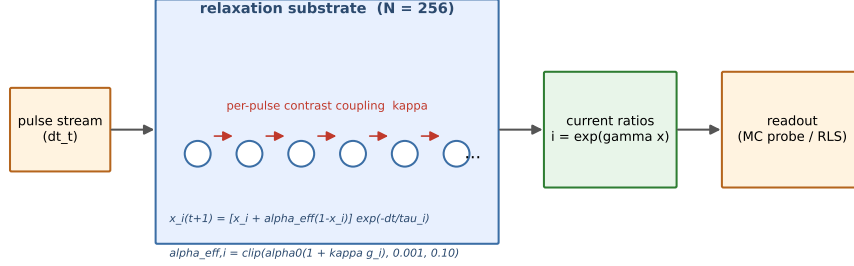


Figure 1: Substrate schematic: pulse stream,  $N = 256$  relaxation units with log-normal  $\tau$  spectrum, per-pulse contrast coupling (red arrows,  $\kappa$ ), and the readout probe.

and the observable current ratio is  $i_i = e^{\gamma x_i}$  with  $\gamma = \ln 100$  (a 100:1 ON/OFF ratio at  $I_{\text{HRS}} = 50$  fA). The trap time constants are drawn from a log-normal distribution with median  $\tau_0 = 174 \mu\text{s}$  (from  $E_a = 0.55$  eV,  $\nu = 10^{13} \text{ s}^{-1}$ ,  $T = 300$  K) and CV = 0.20:  $\tau_i \sim \text{Lognormal}(\mu, \sigma)$  with  $\sigma^2 = \ln(1 + \text{CV}^2)$ ; the log-normal form follows from a Gaussian spread of activation energies (Appendix A.1).

## 2.2 Per-pulse topology coupling

The injection coefficient of unit  $i$  at pulse  $t$  is modulated by a topology-dependent contrast between its own current ratio and a weighted mean of its neighbors' current ratios:

$$\alpha_{\text{eff},i}(t) = \text{clip}\left(\alpha_0(1 + \kappa g_i(t)), \alpha_{\min}, \alpha_{\max}\right), \quad (2)$$

with two contrast conventions — *negative feedback* (mode 1):  $g_i = (\bar{i}_{\mathcal{N}(i)} - i_i)/i_i$ ; and *positive feedback* (mode 2):  $g_i = (i_i - \bar{i}_{\mathcal{N}(i)})/\bar{i}_{\mathcal{N}(i)}$  — where  $\bar{i}_{\mathcal{N}(i)}$  is the weighted mean of neighbor current ratios (row-normalized weights). The base injection coefficient is  $\alpha_0 = 0.02$ ; the physical clip range  $[\alpha_{\min}, \alpha_{\max}] = [0.001, 0.10]$  bounds the injection coefficient. A control condition replaces the contrast coupling with an additive state drive  $x_i \leftarrow x_i + \kappa \bar{x}_{\mathcal{N}(i)}$  after the standard update (the non-physical, “textbook ESN” style). All coupling is computed two-pass (contrasts from the pre-update state, then all units updated), making the step order-independent (Appendix A.2). At  $\kappa = 0$  the model reduces exactly to the parallel array of [4] (verified to machine precision). Fig. 1 summarizes the full pipeline: the pulse stream, the relaxation array with its log-normal  $\tau$  spectrum, the per-pulse contrast coupling, and the linear readout probe.

Five topologies over the  $N = 256$  units are used: **ring\_bidir** (each unit watches its two neighbors), **ring\_unidir** (each unit watches its predecessor), **hub\_star** (satellites watch a fixed hub), **lateral\_ring** (distance-decayed neighbors within radius 4), and **random\_graph** (Erdős–Rényi, mean degree 8, fixed structure across the sweep; graph-to-graph variability of this fixed-instance choice is quantified in Section 4.1). The drive is an i.i.d. uniform interval stream  $\Delta t_t \sim U[2, 20] \mu\text{s}$ , the fast-drive regime in which the nominal memory window is  $N_{\text{eff}} = \tau_0/\bar{\Delta t} \approx 16$  pulses. This i.i.d.

drive differs from the prior parallel-array benchmark protocol, which keyed class information into two *fixed* intervals (20 and 200  $\mu\text{s}$ ) [4]; the interval notation of the two protocols is unrelated.

### 3 Characterization methods

**Finite-time Lyapunov exponent.** For a driven system the relevant quantity is sensitivity to initial conditions under identical drive. We evolve a twin trajectory perturbed by  $\epsilon = 10^{-8}$  per unit, renormalize the separation to  $\epsilon\sqrt{N}$  every 10 pulses (Benettin scheme [24]), and report  $\lambda = \langle \ln(d/\epsilon\sqrt{N}) \rangle$  per pulse, averaged over the full stream (detailed iteration steps in Appendix A.4).

**Held-out memory capacity.** Following Jaeger, the linear memory of lag  $k$  is the squared Pearson correlation between the ridge-decoded value of the driving interval  $\Delta t_{t-k}$  and its true value, decoded from the current-ratio observables at time  $t$ . We fit the ridge readout [25] (regularization  $\lambda_r = 1$ , features standardized with training-segment statistics) on the first 70% of the collected stream, hold out a  $k_{\text{max}} = 50$ -step leakage buffer, and evaluate on the last 30%;  $\text{MC}_{\text{total}} = \sum_{k=1}^{50} r_k^2$ . The held-out protocol is essential: the in-sample convention inflates memory in the chaotic regime by 7–19 $\times$  relative to held-out evaluation (Section 4.4). The estimator derivation and the linear-reservoir closed form that anchors the theory are given in Appendix A.5.

**Input separation.** The RMS distance between the state trajectories produced by two different i.i.d. drives from the same initial condition, averaged over the collected window — a proxy for how far different input histories push the state.

**Activity proxy.** The running standard deviation of the population-mean current ratio over a 200-pulse window. It is not reported as a result below; it enters only as the quantity on which the fixed-activity-target baseline of Section 6 is expressed.

**Reproducibility details.** All phase-diagram runs use  $N = 256$  units,  $\text{CV} = 0.20$ , and 10 independent seeds; the  $\tau$  draws are *paired* across  $\kappa$  within a topology (the same seed index maps to the same  $\tau$  vector at every  $\kappa$ , so  $\kappa$  acts as the dose and the substrate draw as the paired block), and the random-graph structure is fixed across the entire sweep. Every trajectory starts from  $x_i = 0$  and is *preprogrammed* with 50 write pulses (width 1  $\mu\text{s}$ , interval 2  $\mu\text{s}$ ) at the base injection coefficient  $\alpha_0 = 0.02$ , which sets the initial occupancy distribution; the first 200 pulses of each 1200-pulse trajectory are discarded as washout before any statistics are collected. The FTLE twin uses  $\epsilon = 10^{-8}$  per unit with renormalization every 10 pulses. The disturbance experiments of Section 6 use the same substrate (10 seeds) driven by a discretized Mackey–Glass series ( $x_{t+1} = (1-b)x_t + a x_{t-\tau}/(1+x_{t-\tau}^{10})$ ) with  $a = 0.2$ ,  $b = 0.1$ ,  $\tau_{\text{MG}} = 17$  steps; 300-step warmup discarded; values normalized and mapped monotonically onto intervals in  $[2, 20] \mu\text{s}$ ; the target is the next series value), read out by an online recursive least squares rule (forgetting factor 0.999, initial covariance 1, regularization  $10^{-4}$ ,  $N + 1$  features). The homeostat updates every 1000 pulses from a 400-pulse Benettin pair, with feedback gain  $\eta = 3$  and regulatory bounds  $\kappa \in [1, 60]$ .

### 4 Phase diagram

All quantities below are means over 10 independent seeds (paired  $\tau$  draws across  $\kappa$ ); the full table is committed as data/substrate\_phase\_diagram\_v2.csv. Table 1 summarizes the per-topology behavior, and Fig. 2 traces the full  $\kappa$ -dependence of the finite-time Lyapunov exponent, the held-out and in-sample MC, and the clip fraction across the topology families.

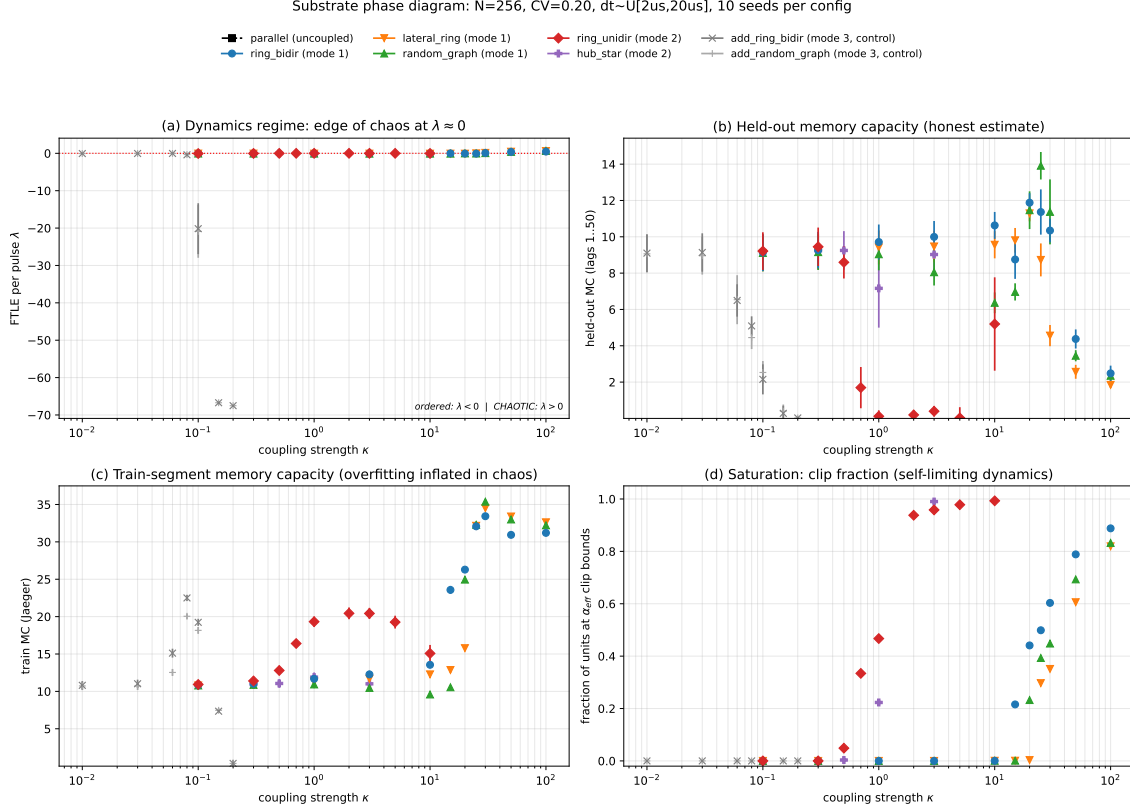


Figure 2: Phase diagram:  $\kappa$  vs. finite-time Lyapunov exponent, held-out and in-sample MC, and clip fraction across topology families.

#### 4.1 Negative-feedback family: stabilization then a sharp transition

For the mode-1 (negative-feedback) topologies, weak-to-moderate coupling *stabilizes*:  $\lambda$  becomes more negative ( $-0.065 \rightarrow -0.090$  as  $\kappa$  grows to 10 for the ring) and the mean absolute contrast  $|g|$  shrinks ( $0.15 \rightarrow 0.01$ ), i.e., the coupling actively homogenizes neighboring states. Memory rises modestly in this regime (ring.bidir held-out MC  $9.1 \rightarrow 10.6$  at  $\kappa = 10$ ). At  $\kappa \in (25, 30)$  the system crosses a sharp order–chaos transition ( $\lambda$  jumps from  $\approx -0.05$  to  $> 0$ ) for all three mode-1 topologies; the transition is accompanied by the onset of  $\alpha_{\text{eff}}$  clipping (clip fraction 0.3–0.6), i.e., the physical bounds start to bind. **The held-out memory capacity peaks in the last ordered configuration before the transition** (peak positions are quoted at the coarse grid resolution  $\Delta\kappa = 5$ ; the fine sweep of Section 4.2 confirms that each peak precedes its topology’s crossing): random\_graph at  $\kappa = 25$  reaches MC = 13.91 vs. 9.07 uncoupled (+53%), ring.bidir at  $\kappa = 20$  +30%, lateral\_ring at  $\kappa = 20$  +24%. The fixed-instance design keeps graph-sample noise out of the topology comparison; a follow-up sweep over 9 independent Erdős–Rényi instances (data/s36\_topo\_instance\_variability\_v1.csv) bounds it: peak held-out MC  $13.47 \pm 0.57$  (range 12.80–14.35), i.e.  $+48.5\% \pm 6.3\%$  over uncoupled, with the peak at  $\kappa = 25$  (7/9 instances) or  $\kappa = 30$  (2/9); the fixed instance of Table 1 (13.91, +53%) lies inside this envelope.

Table 1: Per-topology summary of the  $\kappa$  sweep (held-out MC; mean  $\pm$  s.d. over 10 seeds). Peak gains are relative to the uncoupled baseline MC =  $9.07 \pm 1.65$ .

| Topology     | Mode     | Transition / behavior                                                         | Peak MC (vs. 9.07)                          |
|--------------|----------|-------------------------------------------------------------------------------|---------------------------------------------|
| ring_bidir   | negative | order-chaos at $\kappa^* \in (25, 30)$                                        | $11.88 \pm 0.84$ (+30%)<br>at $\kappa = 20$ |
| lateral_ring | negative | order-chaos at $\kappa^* \in (25, 30)$                                        | $11.29 \pm 0.70$ (+24%)<br>at $\kappa = 20$ |
| random_graph | negative | order-chaos at $\kappa^* \in (25, 30)$                                        | $13.91 \pm 1.22$ (+53%)<br>at $\kappa = 25$ |
| ring_unidir  | positive | self-limited criticality ( $\lambda \approx -0.002$ ),<br>$\kappa \in [1, 5]$ | 0.12–0.39*                                  |
| hub_star     | positive | frozen (clip fraction 1.0)                                                    | none                                        |

\*no held-out linear memory (in-sample MC 18–20).

## 4.2 Transition mechanism: coupling versus the physical clip

The sharp transition at  $\kappa^*$  coincides with the onset of  $\alpha_{\text{eff}}$  clipping (clip fraction 0.3–0.6), which raises the question of whether the chaos is produced by the topology coupling or by the physical bounds. A follow-up sweep (data/s27\_clip\_kappa\_fine\_v1.csv;  $\kappa \in [20, 30]$  at unit resolution  $\times \alpha_{\text{max}} \in \{0.1, 0.2, 0.5\}$ , 10 seeds) answers it. First, the fine grid pins the mean-FTLE crossing at  $\kappa^* = 25.3$  (lateral\_ring), 27.4 (ring\_bidir), and 27.9 (random\_graph), all inside the (25, 30) bracket above — the transition is sharp at unit resolution, not a grid artifact. Second, widening the clip five-fold ( $\alpha_{\text{max}} = 0.5$ ) does not move the transition for the two topologies with the largest memory gains:  $\kappa^* = 25.3 \rightarrow 25.4$  (lateral\_ring; shift +0.05) and  $27.9 \rightarrow 27.7$  (random\_graph; shift –0.21). The order-chaos transition is therefore *coupling-driven*: the clip is a consequence of the large contrasts it bounds, not their cause. ring\_bidir is the one exception — its crossing moves +3.4 under the wider clip, so the bound contributes to that topology’s transition onset. A secondary result: widening the clip raises the ring\_bidir memory peak from 11.9 to 17.7 ( $\alpha_{\text{max}} = 0.2$ ), showing that for ring-like topologies the physical clip currently caps held-out memory — a fabrication-tunable trade.

## 4.3 Positive-feedback family: self-limited criticality, separation without linear memory

The mode-2 (positive-feedback) topologies behave very differently. The **unidirectional ring** self-limits at criticality:  $\lambda$  approaches  $\approx -0.002$  (within our FTLE resolution of zero) over a half-decade of coupling strength  $\kappa \in [1, 5]$ , held there by the  $\alpha_{\text{eff}}$  clip (clip fraction 0.5–1.0) interacting with the  $(1 - x)$  saturation; beyond  $\kappa = 5$  the ordered regime reasserts itself ( $\lambda \approx -0.026$  at  $\kappa = 10$ ). This regime shows the *largest* input separation of the entire study (inter-stream RMS 0.054–0.085 vs. 0.022 baseline) but *zero* held-out linear memory (MC 0.12–0.39, despite in-sample MC of 18–20). The exponential current nonlinearity makes the saturated critical states information-rich but linearly undecodable — a clean demonstration of the separation–memory trade-off: separation-rich critical states require nonlinear readout to be exploited. The **hub\_star** topology freezes instead: with no feedback loop through the hub, satellites saturate into bistable above/below-hub states (clip fraction 1.0) and memory never exceeds the baseline.

#### 4.4 Chaos destroys memory; in-sample MC is misleading

In the chaotic regime ( $\kappa = 50, 100$ ), the held-out MC collapses to 1.85–4.4 (a 3–7 $\times$  drop from the peak) — a sharp contrast to the successful reservoir prediction of externally driven chaotic systems [26, 27], where the drive, not the internal dynamics, supplies the input trace. The in-sample (train-segment) MC, by contrast, *explodes* to 30–35 — an overfitting artifact of ridge decoding the huge-amplitude chaotic state; the held-out protocol reveals the true destruction of memory. We therefore report held-out MC throughout and recommend the practice to the field.

#### 4.5 The additive control: why physics constraints matter

The additive state-coupling control (mode 3, textbook ESN style) reaches its best in-sample MC (17–22) near its saturation threshold ( $\kappa \approx 0.08$ –0.1) and then dies: the state pins at the saturation boundary and the FTLE estimate collapses numerically (a signature of degenerate dynamics). Its held-out MC never exceeds the uncoupled baseline meaningfully (max 9.1), and the operating window is a narrow band before saturation. The physics-constrained contrast coupling, by contrast, remains well-behaved across three decades of  $\kappa$  and achieves the memory gains above — the physical clip acts as a built-in stability regulator.

**Validity domain of  $\kappa$ .** The phase structure above is established on the swept grids —  $\kappa \in [0.1, 100]$  for the contrast-coupled families and  $\kappa \in [0.01, 0.2]$  for the additive control — at a single clip range  $[\alpha_{\min}, \alpha_{\max}] = [0.001, 0.10]$  and  $\text{CV} = 0.20$ ; the clip-ablation sweep of Section 4.2 additionally covers  $\kappa \in [20, 30]$  at unit resolution for  $\alpha_{\max} \in \{0.1, 0.2, 0.5\}$ , and the regulator of Section 6 operates within the regulatory bounds  $\kappa \in [1, 60]$ . Between coarse grid points the transition location is resolved only to its bracketing interval, and outside the swept or clipped ranges — or for spectrum parameters other than those swept in Supplementary Note 1 — the diagram should not be extrapolated.

### 5 Multi-timescale forgetting theory

#### 5.1 The kernel

The linear memory of a single trap for an input at lag  $t$  decays as  $e^{-t/\tau}$ . The substrate’s population forgetting kernel is the average over the log-normal spectrum:

$$M(t) = \int_0^\infty p(\tau) e^{-t/\tau} d\tau, \quad p(\tau) = \text{Lognormal}(\tau_0, \text{CV}). \quad (3)$$

We evaluate  $M(t)$  by Gauss–Hermite quadrature (160 nodes, exact to machine precision; derivation in Appendix A.3). Three properties follow (Fig. 3):

1. **The  $1/e$  horizon is median-pinned.**  $M(\tau_0/\bar{\Delta}t) \approx e^{-1}$  — within a few percent for  $\text{CV} \leq 0.5$  and within  $\sim 25\%$  at  $\text{CV} = 1.0$ ; the horizon is 12–16 pulses for  $\text{CV} \in [0.02, 1.0]$ , in agreement with the nominal window  $N_{\text{eff}} \approx 16$ . The *median* time constant sets where memory is; the width sets how it decays.
2. **The tail steepens with narrowing spectrum.** The log-log slope  $d \ln M / d \ln t$ , fitted over the long-lag window  $t \in [5,000, 20,000]$  pulses (Appendix A.3), is  $-60.6$  at  $\text{CV} = 0.2$  and rises to  $-7.1$  at  $\text{CV} = 1.0$  (and  $-\infty$  in the  $\text{CV} \rightarrow 0$  single-exponential limit). A wider trap spread yields a heavier tail — slower forgetting of old information.

Log-normal tau forgetting kernel: physics-designed decay curve

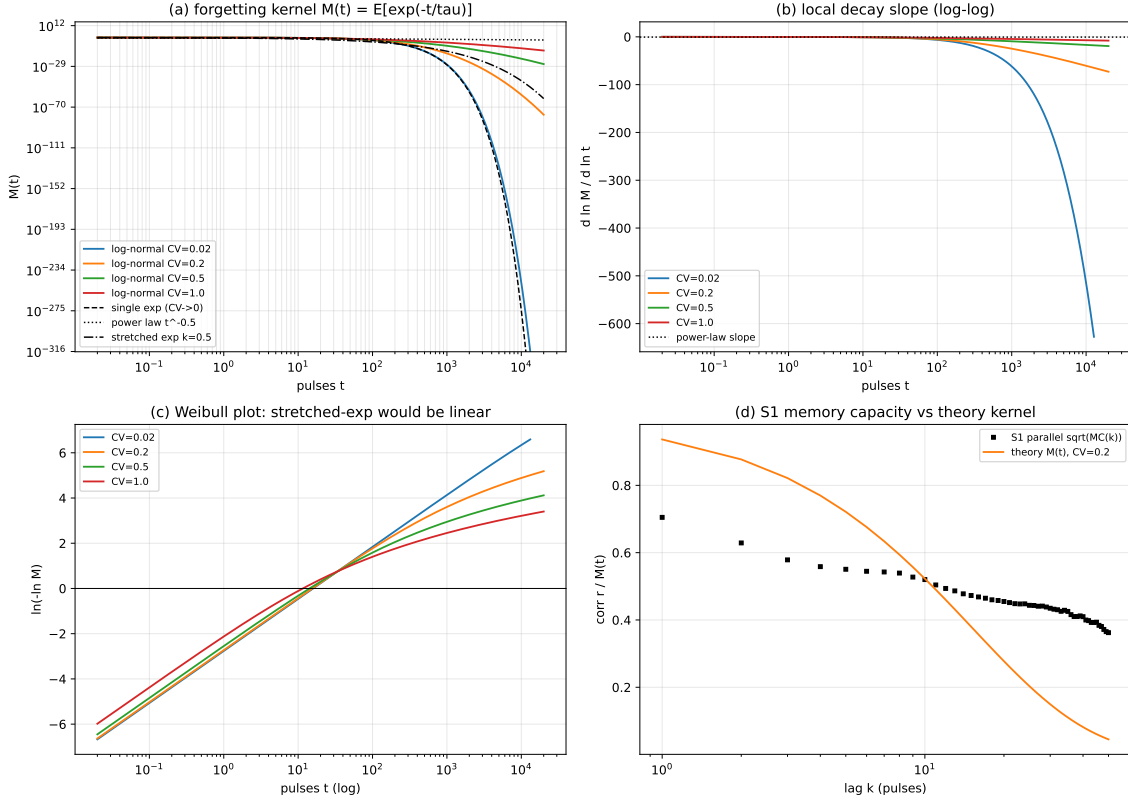


Figure 3: Forgetting kernel:  $M(t)$  curves over CV, tail slopes, and the overlay of the measured  $\sqrt{\text{MC}(k)}$  on  $M(k\Delta t)$  (Pearson  $r = 0.97$ ).

- Comparison to cascade ideals.** The log-Gaussian tail  $\ln M(t) \sim -(\ln t - \mu)^2 / (2\sigma^2)$  is *slower* than a single exponential but *steeper* than the near power-law retention of multi-timescale cascade models [21, 22]. Reaching power-law-like retention would require an unrealistically wide spectrum; the physical substrate therefore occupies a specific point in the forgetting design space set by CV.

## 5.2 Validation against the measured memory capacity

The measured parallel-substrate curve  $\sqrt{\text{MC}(k)}$  (held-out) tracks  $M(k \cdot \bar{\Delta} t)$  at  $\text{CV} = 0.20$  with **Pearson**  $r = 0.97$  (data/forgetting\_curve\_theory\_overlay\_v1.json); at lag 10 the two agree almost exactly (0.520 vs. 0.523), with systematic damping at short lags (readout regression attenuation for a random drive). The committed per-lag curves quantify the residuals: the measured curve sits  $\approx 20\%$  below the kernel for lags  $k \leq 9$  (the readout damping), crosses it at  $k \approx 10$ , and sits above it at long lags (measured 0.36–0.42 vs. kernel 0.04–0.10 for  $k = 36$ –50), so the tail of the driven, readout-limited curve is heavier than the single-unit kernel and the kernel should be read as the bulk, not the extreme tail, of the memory. The device-physics spectrum thus predicts the substrate’s empirical memory curve within the model: the kernel and the measured curve share the same log-normal spectrum assumption, so this is a model-internal consistency check rather than a prediction against independent measurement.



**Implication.** CV — a fabrication-controllable spread parameter — is a *design knob for the forgetting curve*. This is the quantitative basis for the “physics as a structured state-space kernel” view: the substrate realizes a specific multi-timescale kernel with a tunable tail.

**The coupled-substrate caveat (task-level CV sweep).** The kernel theory is single-unit; at the task level the CV knob’s direction is operating-regime-dependent — uncoupled operation follows the theory (wider CV, heavier tail, more retained memory), while coupled near-critical operation prefers narrow spectra. The full sweep ( $CV \in \{0.1, 0.2, 0.4\} \times \{\text{uncoupled, random\_graph } \kappa = 25, \text{ring\_bidir } \kappa = 20\}$ , 10 seeds) is given in Supplementary Note 1 (Fig. S1); the kernel-level stability across  $CV \in [0.02, 1.0]$  is in Fig. 3 and Appendix A.3.

**The kernel survives coupling (shape stability).** The single-unit theory was validated on the parallel substrate; the committed per-lag memory curves let us ask whether the coupled substrate’s curve is still kernel-shaped. The *physical* spectrum (174  $\mu\text{s}$ , CV 0.20) achieves Pearson  $r = 0.91\text{--}0.99$  against  $\sqrt{\text{MC}(k)}$  over  $\kappa \in \{15, \dots, 30\}$  of all three mode-1 topologies ( $r = 0.98$  at random\_graph  $\kappa=25$ ), without any re-fitting (data/kernel\_coupling\_shape\_v1.json; quadrature as in Appendix A.3): the kernel shape survives coupling. The approach to the transition dilates the memory kernel (consistent with the near-critical slowdown of Section 4.1), but single curves cannot separate an effective-median shift from a width change (the fit saturates at wide CV), so we report the shape stability quantitatively and the dilation qualitatively rather than as a precise renormalization map.

## 6 Disturbance robustness and the $\lambda$ -homeostat

### 6.1 The disturbed memory landscape

We subject the random-graph substrate ( $\kappa = 25$ , the nominal optimum) to three abrupt disturbances at  $t = 10\text{k}$  in an online Mackey–Glass forecasting task [28] (recursive least squares, RLS, readout): **temperature drift** (all  $\tau$  scaled by 1.5, i.e. a cooling drift  $\approx 300 \rightarrow 294\text{K}$  under the Arrhenius model; heating would shorten  $\tau$ ), **edge pruning** (40% of coupling edges removed), and **readout noise** ( $\sigma = 0.1 \times$  feature std). The held-out MC- $\kappa$  landscape shifts with the disturbance (means  $\pm$  s.d. over 10 seeds): noise reduces memory everywhere (peak  $10.05 \pm 0.81$  vs.  $13.98 \pm 2.05$  nominal), edge pruning *improves* memory ( $16.53 \pm 2.22$  — removing homogenizing edges), and temperature drift flattens the landscape, retaining substantial memory at larger  $\kappa$  ( $11.01 \pm 1.76$  at  $\kappa = 40$  vs.  $6.95 \pm 0.91$  nominal). (The nominal 13.98 is the disturbance-free peak under this section’s online task protocol; it is distinct from the phase-diagram peak 13.91 of Table 1, which uses the offline ridge protocol of Section 3 on the i.i.d. drive — the 0.5% offset between the two lies well within the seed-level spread of either estimate.) Consequently, any regulator anchored to a *fixed target on the activity proxy of Section 3* (a common homeostatic design) fails: the target’s optimum is not disturbance-invariant (this negative design result motivated the  $\lambda$ -homeostat).

### 6.2 The $\lambda$ -homeostat

The homeostat estimates  $\lambda$  directly: every 1000 pulses it runs a 400-pulse Benettin pair at the current  $\kappa$  (40% computational overhead) and steps

$$\kappa \leftarrow \text{clip}(\kappa + \eta \text{clip}(\lambda_{\text{target}} - \hat{\lambda}, -1, 1), [1, 60]), \quad \lambda_{\text{target}} = -0.02, \quad \eta = 3, \quad (4)$$

the slightly-ordered side of the transition where held-out MC peaks (Section 4.1). We retain  $\lambda_{\text{target}} = -0.02$  in the main results as a deliberately conservative, manually tuned choice made *before* the systematic sweep below; the sweep identifies  $\lambda_{\text{target}} = 0$  as optimal, so the main-result gains

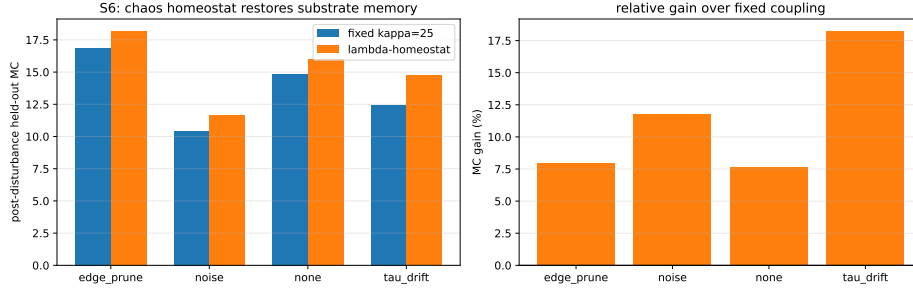


Figure 4:  $\lambda$ -homeostat robustness: post-disturbance held-out MC gains over fixed coupling (Table 2).

should be read as a lower bound on what a retuned target achieves. The regulated system settles at  $\kappa \approx 26$ – $27$  under all conditions and achieves post-disturbance held-out MC gains of  $+7.6\%$  (none),  $+18\%$  ( $\tau$ -drift),  $+7.9\%$  (edge pruning),  $+11.8\%$  (noise) over the fixed coupling (Table 2; Fig. 4), at broadly comparable task-level normalized mean squared error (NMSE; regulated up to  $2\times$  worse on the gain arms, better under readout noise; the RLS readout absorbs task-level differences, and the substrate-level memory improvement is revealed by the readout-independent MC probe). The  $\lambda$ -homeostat is disturbance-type-agnostic because it tracks the dynamical quantity itself rather than a proxy whose optimum shifts.

The estimate is also robust to its own noise: injecting Gaussian error of up to  $\sigma_\lambda = 0.10$  (the typical estimate magnitude is  $0.05$ – $0.1$ ) into every Benettin estimate leaves the settled coupling essentially unchanged (settled  $\kappa$  at the third disturbance round r3:  $28.5$ – $28.7$ , under the S11-chain protocol, which differs from the Mackey–Glass task values of Table 2) and the post-disturbance r3 MC within  $0.16$  of the noiseless value with no significant paired difference at any noise level (two-sided paired  $t$ -test across seeds,  $\alpha = 0.05$ ; data/s32\_ftle\_noise.v1.csv) — the clipped proportional feedback averages out the estimation error over the 1000-pulse interval between updates.

A systematic  $\lambda_{\text{target}}$  sweep ( $\lambda_{\text{target}} \in \{-0.05, -0.02, -0.005, 0\} \times \text{CV} \in \{0.1, 0.2, 0.4\}$ ; 5 seeds, trading per-cell precision for coverage of the 12-cell grid, whereas the single-configuration runs elsewhere in this paper use 10 seeds) was executed under the companion Paper B protocol and is reported as its Table 2 [29] — the values are quoted from that work and are not re-collected under this paper’s phase-diagram protocol. The sweep identifies  $\lambda_{\text{target}}=0$  (the edge of chaos) as optimal: post-disturbance MC peaks near the edge of chaos at every tested CV ( $\lambda_{\text{target}}=0$  optimal at CV=0.1 and 0.4; the  $-0.005$  setting marginally ahead at CV=0.2), with gains of  $+25\%$  (CV=0.1),  $+19\%$  (CV=0.2),  $+5\%$  (CV=0.4) over fixed coupling — updating the manually tuned  $-0.02$  used in the main results.

**Hyperparameter provenance.** The parameters that enter these experiments are of three kinds: measured material properties ( $\tau_0$ , CV); quantities swept in this paper ( $\kappa$  across three decades, Section 4; the clip range, Section 4.2; the task-level CV, Supplementary Note 1); or fixed protocol values inherited from the calibrated prior substrate and the readout conventions ( $\alpha_0$ , the clip range, the preprogramming protocol; ridge  $\lambda_r = 1$  and the 70/30 split;  $\eta = 3$  and the 1000-pulse update period of the homeostat, whose robustness to estimate noise is verified above). No readout or regulator hyperparameter was tuned per condition.

Table 2: Post-disturbance held-out MC of the  $\lambda$ -homeostat (random\_graph substrate, Mackey–Glass online task; mean  $\pm$  s.d. over 10 seeds). Fixed: regulator disabled,  $\kappa = 25$ ; regulated: settled  $\kappa$  (26–27).

| Disturbance   | Fixed MC         | Regulated MC     | Gain   |
|---------------|------------------|------------------|--------|
| none          | $14.87 \pm 2.43$ | $16.01 \pm 2.38$ | +7.6%  |
| $\tau$ -drift | $12.46 \pm 1.74$ | $14.73 \pm 1.88$ | +18%   |
| edge pruning  | $16.87 \pm 1.54$ | $18.21 \pm 1.29$ | +7.9%  |
| readout noise | $10.45 \pm 0.99$ | $11.68 \pm 1.12$ | +11.8% |

## 7 Discussion

**The physical parameter range bounds the accessible computation.** Three material/design quantities jointly define the substrate’s computational envelope: the clip range  $[\alpha_{\min}, \alpha_{\max}]$  (which bounds coupling and, in the positive-feedback family, self-limits the system at criticality), the coupling threshold  $\kappa^*$  (which separates the memory-rich ordered side from the memory-destructive chaotic side), and the trap spectrum (which fixes both the  $1/e$  memory horizon and the forgetting tail). A clip-ablation sweep (Section 4.2) shows that  $\kappa^*$  is invariant to a five-fold widening of the clip for the two highest-gain topologies (random graph and lateral ring), so the threshold is a genuine property of the coupling dynamics rather than a clip artifact; the exception is the bidirectional ring, whose transition shifts by +3.4 under clip widening (Section 4.2), so the clip’s role is secondary and topology-dependent. For the neuromorphic-design question of what this device can compute, the answer is a bounded but substantial region of the memory–chaos plane, fully characterized here.

**Separation without linear memory.** The positive-feedback family reaches true criticality ( $|\lambda| \approx 0$ ) with maximal separation yet zero held-out linear memory. This is a caution for the common practice of tuning reservoirs to the edge of chaos: at criticality the *linear* readout loses the input trace, and the separation must be exploited nonlinearly (a companion work develops structure plasticity from functional connectivity [29]). For linear readouts, the optimal operating point is the *last ordered configuration before the transition* — a slightly-subcritical target ( $\kappa = 20$ –25,  $\lambda \approx -0.05$ ), consistent with where the homeostat parks the system ( $\kappa \approx 26$ –27).

**The forgetting kernel as a bridge between device physics and state-space models.**  $M(t)$  connects device physics to the state-space-model literature: the substrate is a hardware-realizable structured state-space kernel whose decay spectrum is set by fabrication (CV). The  $r = 0.97$  validation shows the connection is quantitative, not metaphorical.

**Spectrum knobs: location versus width.** Across the companion papers three single-parameter spectrum knobs appear — the material CV here, the algorithmic EMA timescale  $\tau_m$  (companion Paper C [30]), and the log-uniform coverage range of the state-space host (companion Paper D [31]). They are not interchangeable, and the distinction is quantitative: for a log-normal spectrum the  $1/e$  horizon is set by the *location* (median  $\tau_0$ :  $M(\tau_0) \approx e^{-1}$  — within a few percent for  $CV \leq 0.5$ , within  $\sim 25\%$  at  $CV = 1.0$ ; numerically 12–16 pulses over  $CV \in [0.02, 1.0]$ ) while the tail weight is set by the *width* (log-log slope  $-60.6$  at  $CV 0.2$  vs.  $-7.1$  at  $CV 1.0$ ).

Paper C’s  $\tau_m$  is a location knob (the synthesized kernel’s horizon scales with  $\tau_m$ , tail shape fixed), and the log-uniform range is a width knob of a different family. The practical corollary: specifying both a horizon and a tail requires two independent knobs, and single-knob tuning moves exactly one observable cross-section — the same decomposition that makes the task-level CV caveat above a re-projection onto a shifted location rather than a contradiction (the effective median dilates under coupling, Section 5).

**Limitations.** All results are numerical (Digital-Model level; no fabricated device); the task-level CV sweep reveals an operating-regime-dependent optimum (Supplementary Note 1); the homeostat’s gains are substrate-level (task-level NMSE is readout-compensated); FTLE estimates are finite-time and resolution-limited near zero.

## 8 Conclusion

A physics-constrained relaxation substrate with per-pulse contrast coupling possesses a well-defined memory–chaos phase diagram: negative-feedback coupling buys 24–53% more held-out memory up to a sharp transition, positive-feedback coupling self-limits at criticality with maximal separation, chaos destroys linear memory, and the log-normal trap spectrum fixes a forgetting kernel that predicts the measured memory curve ( $r = 0.97$ ) with a fabrication-tunable tail. A  $\lambda$ -homeostat restores 7.9–18% of memory after disturbances. These results turn a device-physics spread parameter into a computational design knob and provide the substrate-level theory for the REDEM online-learning architecture (companion paper [29]).

## A Derivation details

### A.1 The trap spectrum is log-normal (Section 2)

A shallow trap escapes by thermally activated emission with rate  $\nu e^{-E/(kT)}$ , so a single trap’s time constant is  $\tau(E) = \nu^{-1} e^{E/(kT)}$ . With median activation energy  $E_a$ ,  $\tau_0 = \nu^{-1} e^{E_a/(kT)}$ ; for  $E_a = 0.55$  eV,  $\nu = 10^{13}$  s $^{-1}$ ,  $T = 300$  K ( $kT \approx 25.85$  meV):  $\tau_0 = 10^{-13} e^{0.55/0.02585}$  s  $\approx 174$   $\mu$ s. If the activation energy varies across devices as  $E_i = E_a + \Delta E_i$  with  $\Delta E_i \sim \mathcal{N}(0, \sigma_E^2)$ , then

$$\ln \tau_i = \ln \nu^{-1} + (E_a + \Delta E_i)/(kT) \sim \mathcal{N}(\ln \tau_0, \sigma_E^2/(kT)^2), \quad (\text{A.1})$$

i.e.,  $\tau_i$  is log-normal with  $\sigma^2 = \sigma_E^2/(kT)^2$  and  $\text{CV}^2 = e^{\sigma^2} - 1$ . The  $\text{CV} = 0.20$  used throughout corresponds to a spread  $\sigma_E = kT \sqrt{\ln(1 + 0.20^2)} \approx 5.1$  meV — a few-meV fabrication spread — which motivates treating CV as a physically plausible design knob rather than a free hyperparameter.

### A.2 Per-pulse update and the order independence of coupling (Section 2)

Over a write pulse of width  $p_w$ , the injection term drives occupancy toward 1 at rate  $\alpha_{\text{eff},i}$ ; over the inter-pulse interval  $\Delta t_t$  both the injected and the pre-existing occupancy relax with the trap time constant. The discrete per-pulse map (Eq. (1)) is the composition of one injection step and one relaxation step:

$$x_i(t+1) = \left[ x_i(t) + \alpha_{\text{eff},i}(t)(1 - x_i(t)) \right] e^{-(p_w + \Delta t_t)/\tau_i}. \quad (\text{A.2})$$

Two-pass order independence: the contrast  $g_i(t)$  is evaluated from the *pre-update* state  $x(t)$  alone (neighbor current ratios at time  $t$ ), and all  $N$  units are then updated simultaneously from the same pre-update state. The map is therefore a well-defined function of the state at  $t$  — a parallel map — and the order in which individual unit updates are computed is irrelevant. At  $\kappa = 0$ ,  $\alpha_{\text{eff},i} = \alpha_0$  for all  $i$  and the coupling term vanishes; the map reduces exactly to the parallel array of [4] (verified to  $\leq 10^{-10}$  in the implementation’s self-test).

### A.3 The forgetting kernel and its quadrature (Section 5)

For a single trap the linear memory of an input at lag  $t$  decays as  $e^{-t/\tau}$ ; averaging over the log-normal spectrum gives Eq. (3). Substituting  $z = (\ln \tau - \mu)/(\sqrt{2}\sigma)$ , with  $d\tau = \sqrt{2}\sigma e^{\mu+\sqrt{2}\sigma z} dz$  and  $p(\tau)d\tau = \pi^{-1/2}e^{-z^2}dz$ :

$$M(t) = \pi^{-1/2} \int_{-\infty}^{\infty} e^{-z^2} \exp(-t e^{-(\mu+\sqrt{2}\sigma z)}) dz \approx \pi^{-1/2} \sum_{i=1}^n w_i \exp(-t e^{-(\mu+\sqrt{2}\sigma z_i)}), \quad (\text{A.3})$$

where  $(z_i, w_i)$  are the  $n$ -point Gauss–Hermite nodes and weights ( $n = 160$  used; the quadrature is exact to machine precision for  $t$  up to  $10^4$  pulses).

**Median pinning of the  $1/e$  horizon.** The integrand peaks at  $\tau^*(t)$  where the saddle-point condition  $t/\tau^* = 1 + (\ln \tau^* - \mu)/\sigma^2$  holds (from  $d/d\tau [\ln p(\tau) - t/\tau] = 0$ ). At  $t = \tau_0$  the solution is  $\tau^* = \tau_0$  up to a weak CV-dependent correction, so  $M(\tau_0) \approx e^{-1}$ : the  $1/e$  horizon is pinned at  $\approx \tau_0/\bar{\Delta}t \approx 16$  pulses regardless of CV (numerically 12–16 pulses for  $\text{CV} \in [0.02, 1.0]$ , Fig. 3).

**Tail asymptotics.** For  $t \gg \tau_0$  the saddle point moves to  $\tau^* \approx t\sigma^2/\ln(t/\tau_0)$ , and  $\ln M(t) \approx -(\ln \tau^* - \mu)^2/(2\sigma^2) - t/\tau^*$ ; dropping the  $t/\tau^*$  term gives the leading-order log–log slope

$$\frac{d \ln M}{d \ln t} \approx -\frac{\ln t - \mu}{\sigma^2}. \quad (\text{A.4})$$

The  $t/\tau^*$  correction is not sub-leading in the sampled window: over  $t \in [5,000, 20,000]$  pulses the exact quadrature slope is  $-60.6$  at  $\text{CV} = 0.2$  ( $\sigma^2 = 0.0392$ ;  $\mu = \ln(\tau_0/\bar{\Delta}t) \approx 2.76$  in pulse units), while Eq. (A.4) gives  $-147$  to  $-182$  — the asymptotic regime is only partially reached (at  $t = 200$  the exact slope is already  $-9.3$ , against the formula’s  $-64.8$ ). The qualitative prediction is confirmed: the slope steepens as the spectrum narrows (quadrature  $-60.6$  at  $\text{CV} = 0.2$  vs.  $-7.1$  at  $\text{CV} = 1.0$ ; Eq. (A.4):  $-147$  to  $-182$  vs.  $-8.3$  to  $-10.3$ ), and  $\text{CV} \rightarrow 0$  makes  $\sigma^2 \rightarrow 0$  and the slope  $\rightarrow -\infty$ : the single-exponential limit.

### A.4 Benettin algorithm (Section 3)

1. Evolve the main trajectory  $x(t)$  under the fixed drive stream for the full collection window.
2. At each renormalization point  $t_m = m T_{\text{ren}}$  ( $T_{\text{ren}} = 10$  pulses), create the twin  $\tilde{x}(t_m) = x(t_m) + \delta$  with  $\delta_i \sim U(-\epsilon, \epsilon)$  per unit,  $\epsilon = 10^{-8}$ .
3. Evolve  $\tilde{x}$  in parallel with  $x$  under the *identical* drive sequence for  $T_{\text{ren}}$  pulses.
4. Measure the separation  $d_m = \|\tilde{x}(t_{m+1}) - x(t_{m+1})\|_2$ , accumulate  $\ln(d_m/(\epsilon\sqrt{N}))$ , and renormalize the twin onto the reference direction:  $\tilde{x} \leftarrow x + (\epsilon\sqrt{N})(\tilde{x} - x)/d_m$ .
5. After  $M$  renormalization blocks,  $\lambda \approx (MT_{\text{ren}})^{-1} \sum_{m=1}^M \ln(d_m/(\epsilon\sqrt{N}))$  per pulse.

The estimate is finite-time and resolution-limited near zero ( $|\lambda| \lesssim 10^{-3}$  is indistinguishable from 0); the perturbation  $\epsilon$  is small enough that the linearized dynamics dominates within one block.

### A.5 Memory-capacity estimator and the linear-reservoir closed form (Section 3)

For lag  $k$  the target is  $y_k(t) = \Delta t_{t-k}$  and the features are the standardized current ratios plus bias,  $\phi_t$ . Ridge regression on the first 70% of the stream solves  $W_k = (\Phi^\top \Phi + \lambda_r I)^{-1} \Phi^\top Y_k$  (targets mean-centered); evaluation is on the last 30% after a  $k_{\text{max}} = 50$ -step leakage buffer, so no evaluation

target references training-segment inputs. The per-lag capacity is the squared multiple correlation  $\text{MC}(k) = r_k^2 = \text{corr}(y_k, \hat{y}_k)^2$ ,  $\text{MC}_{\text{total}} = \sum_{k=1}^{50} r_k^2$ .

*Linear closed form.* Linearize the substrate (clip inactive,  $\kappa$  small):  $x_{t+1} = Ax_t + bu_{t+1}$  with  $u_t = \Delta t_t$  and  $A$  contractive. Then  $x_t = \sum_{j \geq 0} A^j b u_{t-j}$ ; the input at lag  $k$  contributes the component  $A^k b u_{t-k}$ , whose covariance with  $x_t$  is  $c_k = A^k b \text{Var}(u)$ . The best linear predictor achieves the squared multiple correlation

$$\text{MC}(k) = \frac{c_k^\top \Sigma_x^{-1} c_k}{\text{Var}(u)}, \quad \Sigma_x = \text{Var}(x_t) = \sum_{j \geq 0} A^j b b^\top (A^\top)^j \text{Var}(u), \quad (\text{A.5})$$

where the division by the target variance  $\text{Var}(u) = \text{Var}(y_k)$  is required for dimensional consistency:  $c_k$  carries the units of  $xu$ , so  $c_k^\top \Sigma_x^{-1} c_k$  alone is the *explained variance* in units of  $u^2$ , and only after normalization by  $\text{Var}(u)$  does  $\text{MC}(k)$  become the dimensionless squared correlation used throughout. Dambre et al. [20] show the total linear capacity of an  $N$ -dimensional linear system is bounded by the state dimension:  $\sum_k \text{MC}(k) \leq N$ , with equality attained by orthogonally normalized dynamics. The measured nonlinear-substrate curve is estimated with the held-out protocol of Section 3; the closed form anchors the theory at the linear limit.

## Supplementary Material

**Supplementary Note 1. Task-level CV sweep (Fig. S1).** At the task level (held-out MC at the memory-optimal operating points,  $\text{CV} \in \{0.1, 0.2, 0.4\}$ , 10 seeds), the uncoupled substrate follows the single-unit kernel theory: MC rises  $2.54 \pm 0.26 \rightarrow 3.37 \pm 0.33$  (+33%) as CV widens from 0.1 to 0.4 — the heavier tail retains more old lags. The coupled substrate at the near-critical optimum behaves *oppositely*: random\_graph  $\kappa = 25$  MC falls  $12.93 \pm 0.62 \rightarrow 7.33 \pm 1.33$  as CV widens (narrow spectrum best), and ring\_bidir  $\kappa = 20$  falls  $9.55 \pm 0.60 \rightarrow 7.86 \pm 1.85$ . The collective near-critical state prefers homogeneous timescales — heterogeneous relaxation rates disrupt the synchronization of the contrast-feedback loop. The CV knob’s direction therefore depends on the operating regime: wide spectra for parallel (kernel-dominated) operation, narrow spectra for coupled near-critical operation.

## Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Author contributions (CRediT)

**Yaming Hu:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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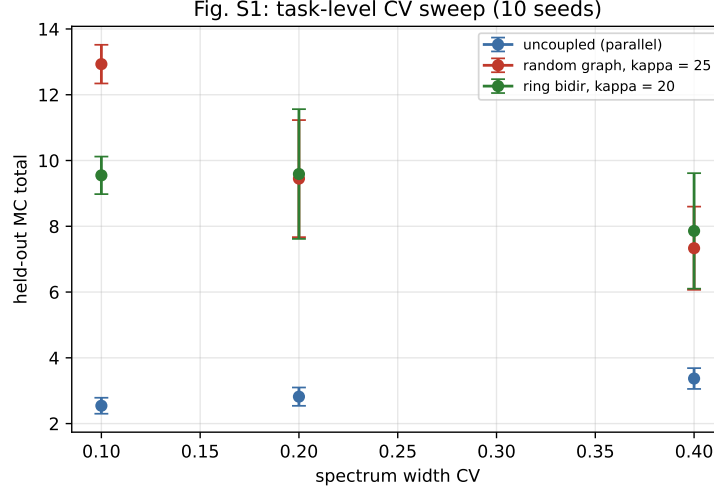


Figure S1: Task-level CV sweep (held-out MC vs. CV, 10 seeds; data/s10\_cv\_sweep.v1.csv).

## Code and data availability

All simulation scripts and generated data are openly available at <https://github.com/huyamingc/REDEM>.

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